Proteome clusters: phenotype, and every result stacked

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Do the people the proteome clustered together differ in muscle size or strength? The phenotype played no part in the clustering, so these tests are not circular. This step reads all five proteome groupings, so it runs after their comparisons, and ends by stacking their results.

For each clustering input and trait: each cluster's median, the Wilcoxon rank-sum test (exact where `wilcox.test()` can compute it) and BH within input. Traits are the % change over training, the composite and volume load, and the T1 and T2 levels. The cross-table with HR/LR describes, with Fisher's exact p; it decides nothing.

Packages

```
library(here)
library(dplyr)
library(tidyr)
library(purrr)
library(tibble)
library(ggplot2)
source(here("00_Setup", "Shared_functions", "analysis_reporting_helpers.R"))
source(here("00_Setup", "Shared_functions", "proteome_inputs.R"))
```

Inputs

```
inputs <- c(
  set_names(
    file.path("02_Groups", paste0("Proteome_", inputs_used), "c_data/clustering.xlsx"),
    paste0("clusters_", inputs_used)
  ),
  phenotype = "00_Setup/Input/phenotype.csv"
)
paths <- map_chr(inputs, here)
stopifnot(file.exists(paths))
assignments <- map(paste0("clusters_", inputs_used), \(name) {
  readxl::read_excel(paths[[name]], "assignments")
}) |>
  list_rbind()
phenotype <- readr::read_csv(paths[["phenotype"]], show_col_types = FALSE)
```

```
pairings <- phenotype_tables(phenotype)$pairings
traits <- bind_rows(
  pivot_longer(pairings$training, -subject, names_to = "trait") |> mutate(kind = "change"),
  pivot_longer(pairings$baseline_level, -subject, names_to = "trait") |> mutate(kind = "T1
level"),
  pivot_longer(pairings$post_level, -subject, names_to = "trait") |> mutate(kind = "T2 level")
)
count(assignments, input, cluster)
```

```
# A tibble: 10 × 3
   input    cluster     n
   <chr>      <dbl> <int>
 1 T1             1     9
 2 T1             2     6
 3 T2             1     7
 4 T2             2     8
 5 T3             1     5
 6 T3             2    10
 7 acute          1     4
 8 acute          2    11
 9 training       1     7
10 training       2     7
```

Tests

```
tests <- assignments |>
  select(input, subject, cluster) |>
  inner_join(traits, by = "subject", relationship = "many-to-many") |>
  filter(!is.na(value)) |>
  nest(.by = c(input, kind, trait)) |>
  mutate(test = map(data, \(x) {
    rank_sum <- suppressWarnings(wilcox.test(value ~ cluster, data = x))
    tibble(
      n_1 = sum(x$cluster == 1), n_2 = sum(x$cluster == 2),
      median_1 = median(x$value[x$cluster == 1]), median_2 = median(x$value[x$cluster == 2]),
      w = unname(rank_sum$statistic), p = rank_sum$p.value
    )
  })) |>
  select(-data) |>
  unnest(test) |>
  mutate(fdr = p.adjust(p, "BH"), .by = input) |>
  arrange(input, p)
head(tests)
```

```
# A tibble: 6 × 10
  input kind     trait           n_1   n_2 median_1 median_2     w      p   fdr
  <chr> <chr>    <chr>         <int> <int>    <dbl>    <dbl> <dbl>  <dbl> <dbl>
1 T1    T2 level fcsa_type2        9     6  6748.    4649.      43 0.0663 0.867
```

```
2 T1    T2 level fcsa_mixed         9     6  6386.    4916.      41 0.113  0.867
3 T1    change   pct_leg_press…     9     6    47.4     15.3     38 0.224  0.867
4 T1    T1 level fcsa_type2         9     6  6097.    5764.      38 0.224  0.867
5 T1    change   pct_mcsa           9     6     5.26    11.2     17 0.272  0.867
6 T1    change   pct_fcsa_type1     9     6    10.7     -9.23    35 0.388  0.867
```

```
hr_lr <- assignments |>
  summarise(
    hr_in_1 = sum(hr_lr == "HR" & cluster == 1), lr_in_1 = sum(hr_lr == "LR" & cluster == 1),
    hr_in_2 = sum(hr_lr == "HR" & cluster == 2), lr_in_2 = sum(hr_lr == "LR" & cluster == 2),
    fisher_p = fisher.test(table(hr_lr, cluster))$p.value,
    .by = input
  )
hr_lr
```

```
# A tibble: 5 × 6
  input    hr_in_1 lr_in_1 hr_in_2 lr_in_2 fisher_p
  <chr>      <int>   <int>   <int>   <int>    <dbl>
1 T1             4       5       3       3    1
2 T2             3       4       4       4    1
3 T3             4       1       3       7    0.119
4 training       3       4       3       4    1
5 acute          3       1       4       7    0.282
```

```
assignments |>
  inner_join(filter(traits, kind == "change"), by = "subject", relationship = "many-to-many") |
>
  ggplot(aes(factor(cluster), value)) +
  geom_boxplot(outlier.shape = NA, width = 0.5) +
  geom_point(position = position_jitter(width = 0.1, seed = 1), size = 1) +
  facet_grid(trait ~ input, scales = "free_y") +
  labs(x = "cluster", y = "value") +
  theme_bw(base_size = 7)
```

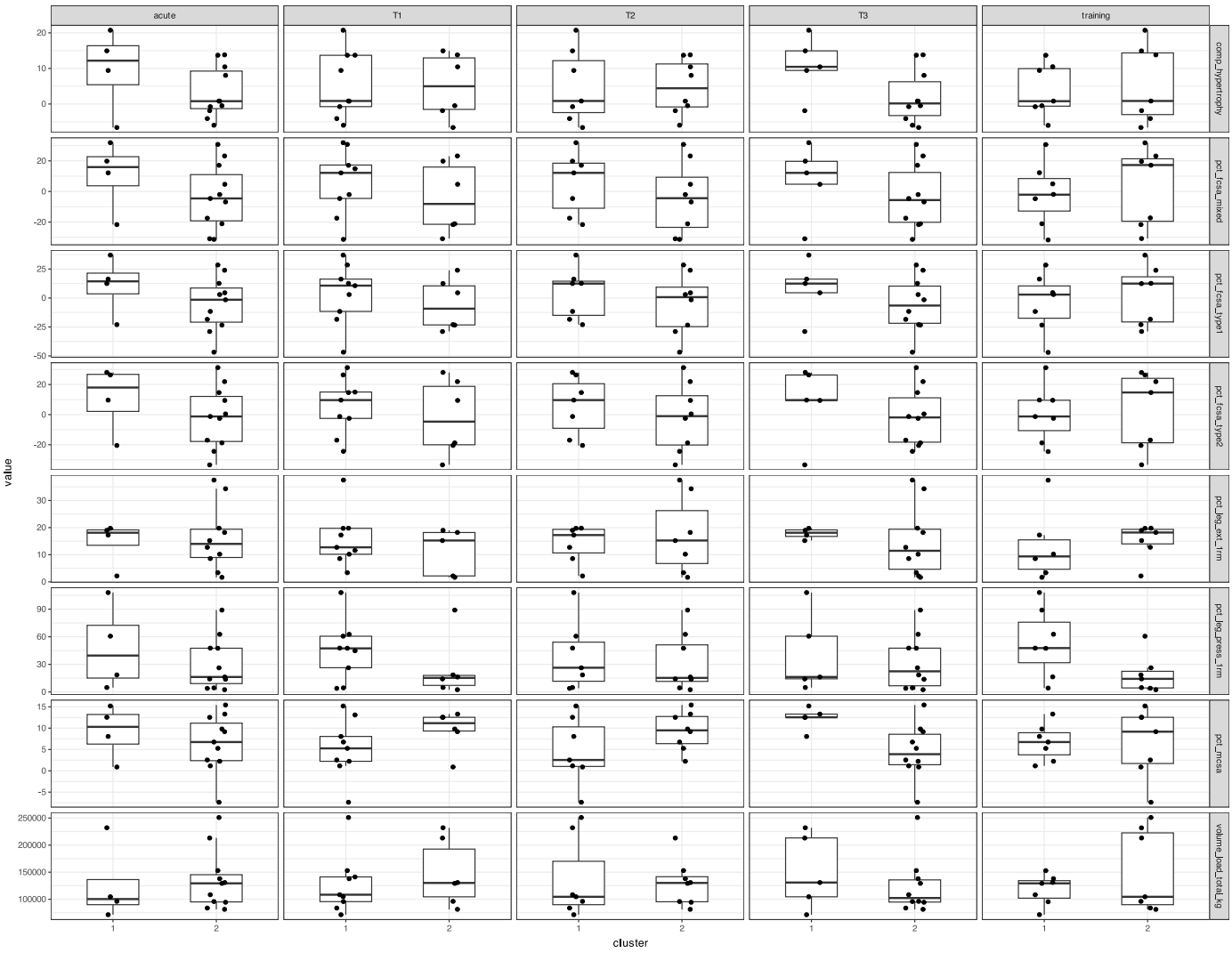


Figure 1: Training response by proteome cluster, per input.

Validation

```
stopifnot(all(tests$fdr >= tests$p), all(tests$n_1 >= 1), all(tests$n_2 >= 1))
```

Summary

100 cluster-against-trait tests over 5 inputs: 4 at p < 0.05, 0 at BH < 0.05.

Save

```
out <- here("03_Group_comparisons", params$grouping, "c_data")
dir.create(out, recursive = TRUE, showWarnings = FALSE)
write_workbook(
  list(tests = tests, hr_lr = hr_lr),
  c(
    "Per input, trait kind and trait: n and median per cluster, Wilcoxon W and p, BH within
```

```
input.",
    "Per input: HR and LR counts in each cluster, Fisher's exact p (description only)."
  ),
  file.path(out, "07_cluster_phenotype.xlsx"), inputs, paths
)
```

Every result, stacked

Each proteome grouping writes its own workbooks. `stack_inputs()` (`proteome_inputs.R`) stacks one workbook's result sheets from all five into `<file>_all_inputs.xlsx`, with an `input` column and a `validity` column: a contrast that uses exactly the time points the clustering was built on is `circular`, one that shares a time point is `caution`, the rest `clean`. The trait associations workbook leaves out its within-group sheet, which would pass the xlsx row limit.

```
walk(
  c(
    "01_design_contrasts.xlsx", "02_protein_contrasts.xlsx", "03_set_contrasts.xlsx",
    "04_module_contrasts.xlsx", "05_protein_classification.xlsx", "05_set_classification.xlsx",
    "05_module_classification.xlsx"
  ),
  stack_inputs
)
stack_inputs("06_trait_associations.xlsx", leave_out = "within_group")
```